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Task:

1.Create a small web api project that fetches the list of pet (dog varieties) from sql server via the code first approach entity framework approach

NOTE this should be completely done in the vs code

Solution:

Initializing the dotnet and creating the Web Api Project in the Visual studio code Terminal

command:

cd D:\C#\ApiUsingVs

dotnet new webapi -n ApiUsingVs

Model Class:

```
namespace ApiUsingVs.Models
{
    public class Pet
    {
        public int Id { get; set; }
        public string Name { get; set; } = string.Empty;
        public string Breed { get; set; } = string.Empty;
    }
}
```

DBcontext Class:

```
using Microsoft.EntityFrameworkCore;
using ApiUsingVs.Models;

namespace ApiUsingVs.Context
{
    public class PetContext : DbContext
    {
        public PetContext(DbContextOptions<PetContext> options) :
base(options) { }

        public DbSet<Pet> Pets { get; set; }
    }
}
```

```
}  
}
```

Controller Class:

```
using Microsoft.AspNetCore.Mvc;  
using Microsoft.EntityFrameworkCore;  
using ApiUsingVs.Context;  
using ApiUsingVs.Models;  
  
namespace ApiUsingVs.Controllers  
{  
    [ApiController]  
    [Route("api/[controller]")]  
    public class PetController : ControllerBase  
    {  
        private readonly PetContext _context;  
  
        public PetController(PetContext context)  
        {  
            _context = context;  
        }  
  
        [HttpGet("GetAll")]  
        public async Task<ActionResult<IEnumerable<Pet>>> GetAll()  
        {  
            return await _context.Pets.ToListAsync();  
        }  
  
        [HttpGet("{id}")]  
        public async Task<ActionResult<Pet>> GetById(int id)  
        {  
            var pet = await _context.Pets.FindAsync(id);  
            if (pet == null) return NotFound();  
            return pet;  
        }  
  
        [HttpPost("Add")]
```

```

        public async Task<ActionResult<Pet>> Add(Pet pet)
        {
            _context.Pets.Add(pet);
            await _context.SaveChangesAsync();
            return Ok(pet);
        }

        [HttpDelete("{id}")]
        public async Task<IActionResult> Delete(int id)
        {
            var pet = await _context.Pets.FindAsync(id);
            if (pet == null) return NotFound();

            _context.Pets.Remove(pet);
            await _context.SaveChangesAsync();
            return NoContent();
        }
    }
}

```

Program Class:

```

using ApiUsingVs.Context;
using Microsoft.EntityFrameworkCore;
using Microsoft.OpenApi.Models;

var builder = WebApplication.CreateBuilder(args);

builder.Services.AddControllers();
builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen(c =>
{
    c.SwaggerDoc("v1", new OpenApiInfo { Title = "Pet API", Version = "v1"
});
});

const string connection = "Data Source=PTPLL614;Initial
Catalog=TrainingDB;Integrated Security=true;TrustServerCertificate=True";

```

```

builder.Services.AddDbContext<PetContext>(options =>
options.UseSqlServer(connection));

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI(c => c.SwaggerEndpoint("/swagger/v1/swagger.json", "Pet
API v1"));

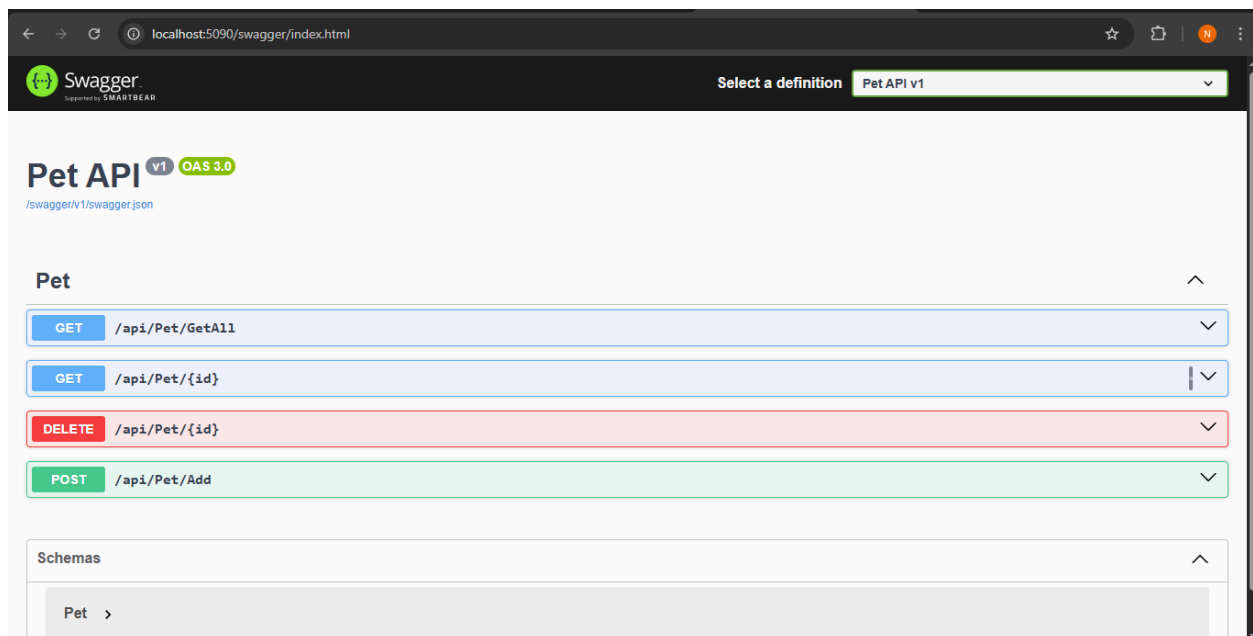
app.UseHttpsRedirection();
app.UseAuthorization();

app.MapControllers();

app.Run();

```

Output :



Add :

DELETE /api/Pet/{id}

POST /api/Pet/Add

Parameters

No parameters

Request body

application/json

```
{
  "id": 0,
  "name": "Rocky",
  "breed": "Labrador"
}
```

Execute Clear

Curl

```
curl -X 'POST' \
  'http://localhost:5090/api/Pet/Add' \
  -H 'accept: text/plain' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 0,
    "name": "Rocky",
    "breed": "Labrador"
  }'
```

Request URL

http://localhost:5090/api/Pet/Add

Server response

Code Details

200

Response body

```
{
  "id": 4,
  "name": "Rocky",
  "breed": "Labrador"
}
```

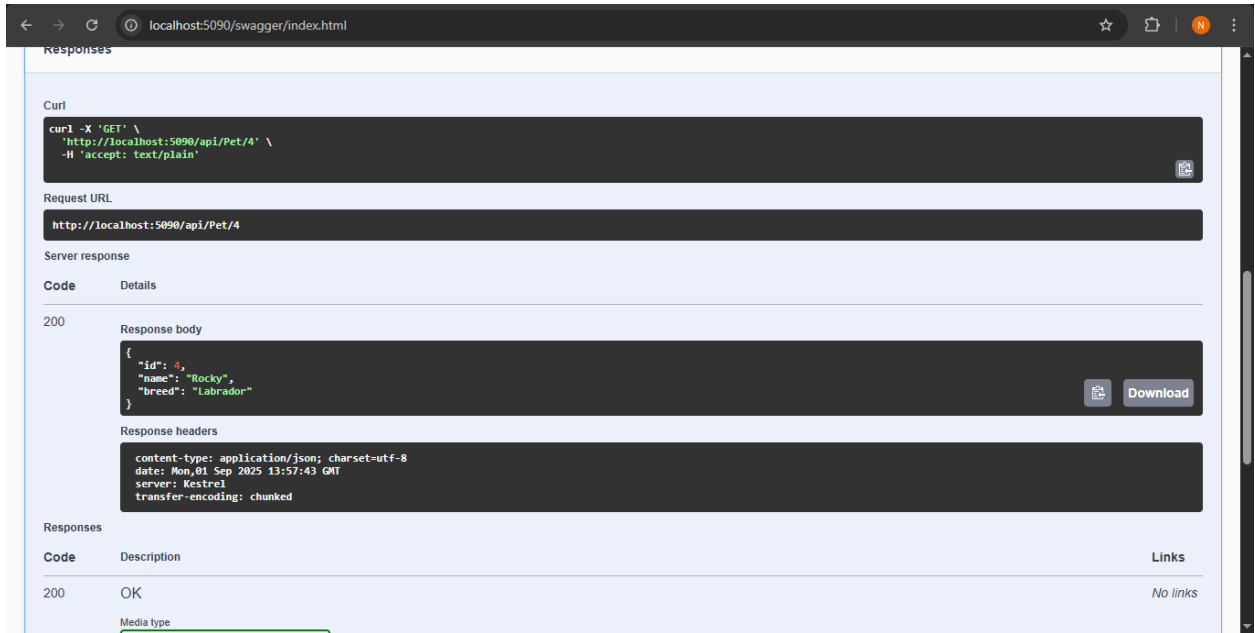
Response headers

```
content-type: application/json; charset=utf-8
date: Mon, 01 Sep 2025 13:57:02 GMT
server: Kestrel
transfer-encoding: chunked
```

Responses

Code	Description	Links
200	OK	No links

Get By ID:



A screenshot of the Swagger UI interface in a web browser. The browser's address bar shows the URL `localhost:5090/swagger/index.html`. The page title is "Responses". The "Curl" section contains the command:

```
curl -X 'GET' \
'http://localhost:5090/api/Pet/4' \
-H 'accept: text/plain'
```

. The "Request URL" section shows `http://localhost:5090/api/Pet/4`. The "Server response" section displays a 200 status code. The "Response body" is a JSON object:

```
{
  "id": 4,
  "name": "Rocky",
  "breed": "Labrador"
}
```

. The "Response headers" section shows:

```
content-type: application/json; charset=utf-8
date: Mon, 01 Sep 2025 13:57:43 GMT
server: Kestrel
transfer-encoding: chunked
```

. At the bottom, a table lists the response with a 200 status code and the description "OK".

Responses

Curl

```
curl -X 'GET' \
'http://localhost:5090/api/Pet/4' \
-H 'accept: text/plain'
```

Request URL

```
http://localhost:5090/api/Pet/4
```

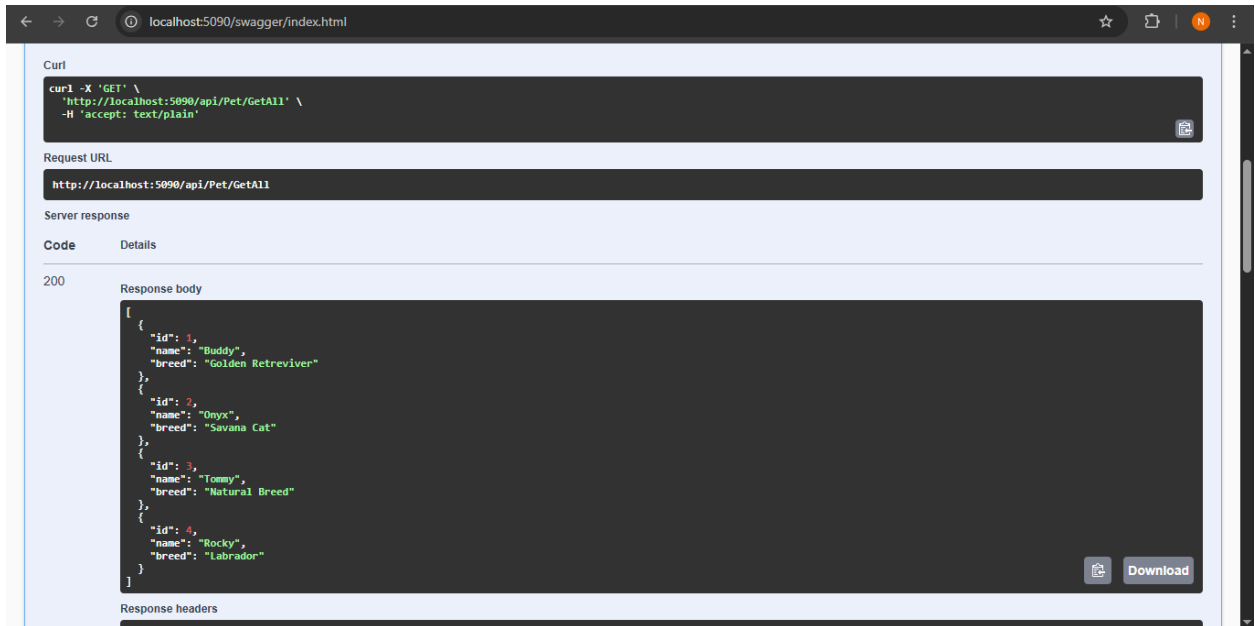
Server response

Code	Details
200	Response body <pre>{ "id": 4, "name": "Rocky", "breed": "Labrador" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Mon, 01 Sep 2025 13:57:43 GMT server: Kestrel transfer-encoding: chunked</pre>

Responses

Code	Description	Links
200	OK	No links

Get All :



A screenshot of the Swagger UI interface in a web browser. The browser's address bar shows the URL `localhost:5090/swagger/index.html`. The page title is "Responses". The "Curl" section contains the command:

```
curl -X 'GET' \
'http://localhost:5090/api/Pet/GetAll' \
-H 'accept: text/plain'
```

. The "Request URL" section shows `http://localhost:5090/api/Pet/GetAll`. The "Server response" section displays a 200 status code. The "Response body" is a JSON array of four pet objects:

```
[
  {
    "id": 1,
    "name": "Buddy",
    "breed": "Golden Retriever"
  },
  {
    "id": 2,
    "name": "Onyx",
    "breed": "Savanna Cat"
  },
  {
    "id": 3,
    "name": "Tommy",
    "breed": "Natural Breed"
  },
  {
    "id": 4,
    "name": "Rocky",
    "breed": "Labrador"
  }
]
```

. The "Response headers" section is empty. At the bottom, a table lists the response with a 200 status code and the description "OK".

Responses

Curl

```
curl -X 'GET' \
'http://localhost:5090/api/Pet/GetAll' \
-H 'accept: text/plain'
```

Request URL

```
http://localhost:5090/api/Pet/GetAll
```

Server response

Code	Details
200	Response body <pre>[{ "id": 1, "name": "Buddy", "breed": "Golden Retriever" }, { "id": 2, "name": "Onyx", "breed": "Savanna Cat" }, { "id": 3, "name": "Tommy", "breed": "Natural Breed" }, { "id": 4, "name": "Rocky", "breed": "Labrador" }]</pre> Response headers

Responses

Code	Description	Links
200	OK	No links

Delete by Id:

localhost:5090/swagger/index.html

Responses

Curl

```
curl -X 'DELETE' \
  'http://localhost:5090/api/Pet/5' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:5090/api/Pet/5
```

Server response

Code	Details
200	Response body <pre>Deleted Successfully</pre> Response headers <pre>content-type: text/plain; charset=utf-8 date: Mon, 01 Sep 2025 14:01:53 GMT server: Kestrel transfer-encoding: chunked</pre>

Responses

Code	Description	Links
200	OK	No links